SnapCare

16TH E-ICON WORLD CONTEST · TEAM H1

A prescription you never have to read.

Namyunsu Nam, ChungNam Samsung Academy
Republic of Korea & Republic of Moldova · hosted by the Ministry of Education

739M
adults worldwide lack basic literacy skills

17/108
pictograms that met the ANSI comprehension threshold

78
morphemes the engine composes from, rather than selects

39
experts in 22 countries answered our survey

Try it now
`snapcare.fly.dev`
Runs in the browser. No install, no account. Sample prescriptions are built in, and it keeps working offline once loaded.

1 The problem

A prescription is written to be read. From the moment it is handed over everything depends on the patient's ability to read it, and for a large population that ability is absent or partial. In Korea 39.6% of adults do not reach a sufficient level of health literacy; among those aged 65 and over it is **59.7%**.

Removing text is necessary but not sufficient. Pictures can fail to be read as well: in the Philippines only **17 of 108** pictograms reached the ANSI Z535.3 threshold among 52 adults [1].

Existing sets are **lists of pictures**. A set of *n* drawings can say *n* things; an instruction not drawn in advance cannot be said at all.

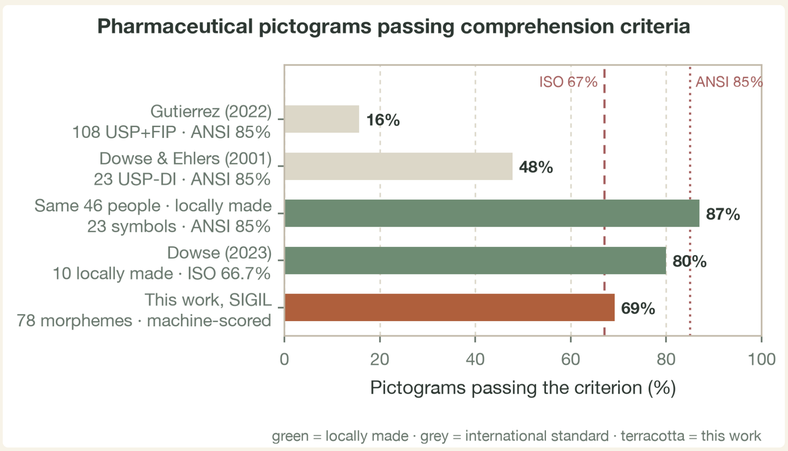


Figure 1. Comprehension against the ISO and ANSI criteria. **Locally made sets pass; the international sets do not.**

2 What must be true

At no point on the path to a judgement is the user required to read or type.

Two channels are laid so that neither leans on the other. The pictures alone, or the speech alone, must reach the same instruction. Text stays on screen for those who can use it, but the instruction has to survive without it.



Figure 2. Either channel on its own has to arrive at the same place.

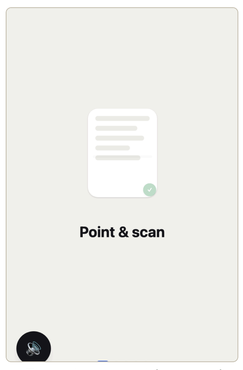
3 How it works

1. **Photograph.** The user points the camera at a prescription. Nothing is typed and no object has to be chosen first.
2. **Read on the device.** Text recognition and form parsing run in the browser. Drug, dose, frequency, timing, duration, storage and contraindications are extracted; the photograph never leaves the device.
3. **Compose the pictures.** SIGIL builds each frame on the spot from morphemes, so an instruction nobody drew in advance can still be shown.
4. **Rebuild the sentence.** VOCA keeps only what the patient needs; the verb and the body site follow from the dosage form.

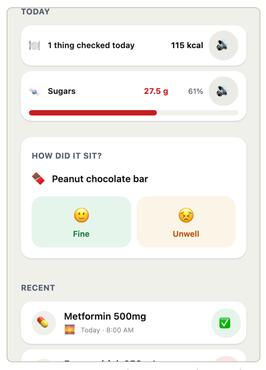
No account	No upload	Offline
nothing to register	the photograph stays put	verified, server off

The three layers of delivery **overlap rather than substitute**. Any one of them failing still leaves the instruction standing, which is why none of them is allowed to be the only carrier of a warning.

Picture	Speech	Colour + symbol
what to do SIGIL composed pictogram	what that picture means VOCA reconstructed sentence	is this safe now never colour alone



d. First run. Five pictures with speech; skippable, and no field to fill.



e. Home. One capture button; the subject is never chosen by the user.

4 SIGIL – a grammar, not a list

A head stands on its own; a modifier cannot, and goes into a slot inside a head. Five operators apply on top. The 78 were derived **backwards from the distribution of medication error**.

Element	n	Examples
Head	57	pill_bottle, allergen_peanut
Modifier	21	time_morning, storage_fridge
Operator	5	no{}, half{}, twice_daily{}
Morphemes	**78**	**composed, not selected**



Figure 3. Every morpheme the engine holds. Nothing here was drawn for a particular prescription.

5 One prescription, composed

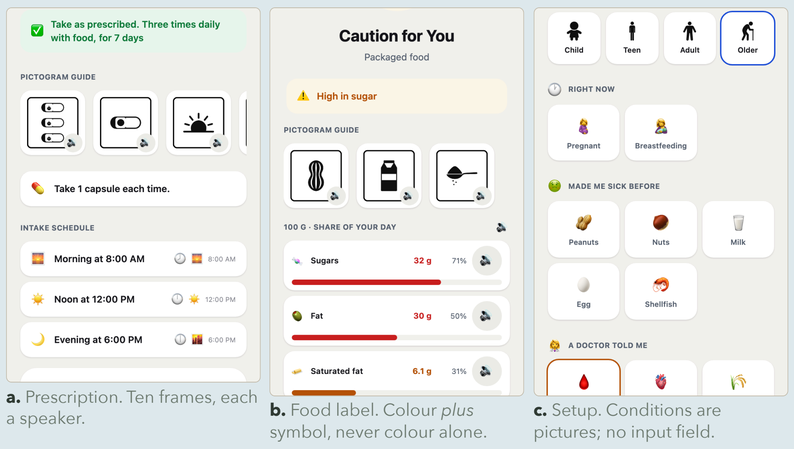
The prescription text is passed through the same path the camera uses. The output below is reproduced verbatim.

```
Amoxicillin capsules 250 mg — three times a
day, 30 min after meals, 1 capsule, 7 days,
refrigerate, not in pregnancy, no alcohol

\three_times_daily{\capsule_strip{\food_bowl}}
\capsule_strip{\time_morning}  \finish_course
\capsule_strip{\time_noon}     \no_alcohol
\capsule_strip{\time_evening}  \state_pregnant
\capsule_strip{\qty_one}       \state_breastfeeding
\capsule_strip{\storage_fridge}
```

Ten frames from one prescription. **Four carry what was buried in the warning box**: finish the course, no alcohol, not in pregnancy, caution while breastfeeding.

The spoken channel does not read that string aloud. The verb follows the dosage form: a tablet is *take*, eye drops are *put in*, and the body site follows from the form.



a. Prescription. Ten frames, each a speaker.

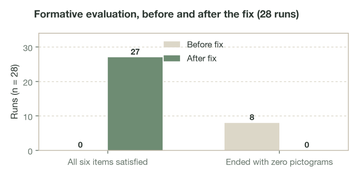
b. Food label. Colour *plus* symbol, never colour alone.

c. Setup. Conditions are pictures; no input field.

Figure 4. What the user sees. Every element on all three screens can be pressed to be read aloud.

6 What we measured

What was measured	Result
Formative runs 4 prescriptions × 7 conditions, 6 items each	**27 / 28** passed all six 0 / 28 before the fix
Semantic check of all 78 morphemes	**54** read as intended the 7 exposed on screen were reworked
Nutrition model, Nutrition5k RGB split	**26.75%** mean MAE n = 507, the deployed web.onnx
Offline, with the server shut down	**Normal** one missing cache entry, found and fixed



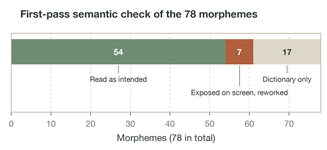


Figure 5. Before and after the fix. **The cause was a missing connection, not a defect in any single feature.**

Figure 6. The 24 read differently are split, not pooled: 7 on screen, reworked; 17 in the dictionary only.

The formative score is kept **per item** rather than reduced to one number. "27 / 28" on its own leaves no way of knowing what failed; the single failure was item D5 under one condition, and it was left uncorrected rather than scored away.

The offline result is not a claim that the network is unnecessary in principle. The server was shut down and the same four screens were driven again, which is how the one missing cache entry was found.

7 We wrote to 1,103 experts



● **13** Pharmacy ● **10** Other ● **9** Nursing ● **4** HCI / design ● **3** Literacy, AAC, industry

Eight domains were surveyed in parallel, a source URL was recorded for every entry and no address was ever invented. Each person received a separate letter carrying an opaque token, so the server stores the token and only our own spreadsheet can say who answered. **Pharmacy and nursing together are 22**, the centre of the fields judged qualified to rule on comprehension.

8 What they told us

A prohibition is not read as a prohibition

> *The prescription read not to use the eyedrops in pregnancy while the pictogram showed a pregnant woman, making the patient assume it is safe to use in pregnancy.*
>
> Prof. Atif Usman, The Islamia University of Bahawalpur, Pakistan

Three respondents on three continents reported this independently. Fixing it uncovered the same accident the other way: `no_alcohol` became `\no{\no_alcohol}` — **a slash through "do not drink" says that drinking is fine**.

The premise itself was questioned

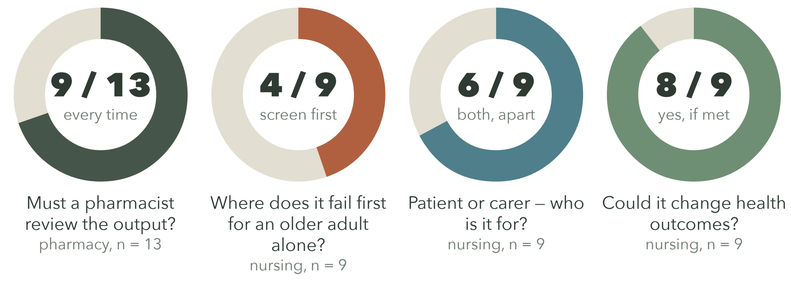
> *To a non-literate person pictograms too can feel like letters. As the saying goes, "the black is the writing and the white is the paper."*
>
> Prof. Heejung Seo, Dong-eui University, Korea

The comprehension studies we cite measured understanding *after* a picture was recognised as a picture. Whether it is recognised as a picture at all was never measured.

Our design could create a new risk

> *Even a medicine that carries risk for pregnant women is sometimes prescribed after weighing the patient's overall condition, and where the patient checks alone they may end up not receiving the drug therapy.*
>
> Prof. So-Young Lee, pharmacist, Chungnam National University, Korea

A personalised verdict cannot beat a personalised prescription.



9 What changed, what we do not claim

- **Contraindication marking was split three ways.** Flags for what is contained gain and lose the slash with the verdict; flags for what not to do are always prohibited.
- `finish_course` **was redrawn.** Three of five cells filled says "not finished yet" rather than "finish it". A directional arrow proposed by a respondent replaced it.
- **No comprehension test with people has been run.** ISO 9186-1 (67%) and ANSI Z535.3 (85%, critical confusions ≤ 5%) must still be confirmed with low-literacy adults.
- **Cultural adaptation is unverified.** Locally produced pictograms passed for 20 of 23 symbols where USP-DI passed for 11 among the same 46 people [3].
- **Experts are not patients.** An expert answering that a patient could probably manage is not evidence that a patient did.
- **The survey's own defects were reported by respondents** and fixed mid-run, including a branch that showed no questions at all.

SDG	Contribution, stated narrowly
3.4	Thresholds derived from the individual's energy requirement
3.8	Reduces the literacy-driven gap after care ends
3.d	Carries course completion as a picture, against resistance

What comes next is the test the survey asked for. ISO 9186-1 with low-literacy adults, scored per morpheme rather than per screen, and the three populations the respondents separated for us recruited apart: total blindness, low vision, low literacy. Until that is run, every comprehension figure on this sheet is a machine's judgement, not a reader's.

[1] Gutierrez, M. M., Patikorn, C. & Anantachoti, P. Evaluation of pharmaceutical pictogram comprehension among adults in the Philippines. *J. Pharm. Policy Pract.* 15:30 (2022).
[2] Berkman, N. D. et al. Low health literacy and health outcomes: an updated systematic review. *Ann. Intern. Med.* 155(2):97–107 (2011).
[3] Dowse, R. & Ehlers, M. S. The evaluation of pharmaceutical pictograms in a low-literate South African population. *Patient Educ. Couns.* 45:87–99 (2001).
[4] Korea Disease Control and Prevention Agency. *KNHANES*, 9th cycle year 2 (2023). [5] ISO 9186-1:2014; ANSI Z535.3.

Reproduction. Every figure here is produced from the repository by one script; the roster, the letter, the item definitions and the response export are held with it.

Acknowledgement. Thirty-nine experts in twenty-two countries answered without compensation, and several reported a defect in the survey itself. Their answers are recorded as given, including the points we cannot answer.